

Q ①: Let $\mathbb{R}$: set of real numbers. Define a binary operation $*$ on $\mathbb{R}^+$ as

$$x * y = \frac{xy}{x+y+1} \quad \text{Verify whether } * \text{ is}$$

Commutative, associative? Whether identity exists?

Q ②: Let G be a group and $a, b, c, x \in G$. Solve simultaneously

$$x^2 a = b x c^{-1} \quad \text{and} \quad a c x = x a c.$$

Q ③: Prove D_8 (with all rotations, reflections, flips) is non-abelian. Draw composition table.

Q ④: Let $\cup(n)$: set of positive integers less than n and relatively prime to n . ($n \geq 1$)
Find order of each element in the group $\cup(15)$, the multiplicative group modulo 15.

Q ⑤: Let G be a commutative group $\{e, a, b, c, ab, bc, ac, abc\}$ whose generators satisfy
 $a^2 = b^2 = c^2 = e$. Write table for G .

Q ⑥: Suppose $a \in G$ and $O(G) = n$ where n is least such that $a^n = e$. Then prove that
 $a^t = e \iff t$ is a multiple of n . ~~Prove~~

Q (7) Express each of the following permutations in S_9 as a product of transpositions:

1. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 7 & 9 & 5 & 3 & 1 & 2 & 4 & 8 & 6 \end{pmatrix}$.

2. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 9 & 8 & 7 & 4 & 3 & 6 & 5 & 1 & 2 \end{pmatrix}$.

Q (8). Prove that the following are equivalence relations on a given set. What are equivalence classes.

(1): $\sim$ on $\mathbb{R}^2$

$$(x_1, y_1) \sim (x_2, y_2) \iff x_1^2 + y_1^2 = x_2^2 + y_2^2$$

(2): $\sim$ on $\mathbb{Z}$, $m \sim n \iff m - n$ is a multiple of 10.

Q (9): Prove the family of all the cosets Ha as a ranges over G , is a partition of G .

Give partition for $(\mathbb{Z}_{24}, +)$ with appropriate subgroup.

Q (10): Prove for a group G , and a subgroup H ,

(1) $aH = bH \iff a^{-1}b \in H$

(2) $|aH| = |bH|$

(3) prove Lagrange's Theorem if G is finite.

Q (11). Let $U(n) = \{x \in \mathbb{Z}^+ \mid x < n, (x, n) = 1\}$.

Find inverse of 9 in $U(10)$

inverse of 3 in $U(10)$

inverse of 17 in $U(20)$

inverse of 11 in $U(20)$

inverse of 13 in $U(20)$.

Q (12): Consider $\mathbb{Z}_n$ ($n \geq 1$).

k is a generator of $\mathbb{Z}_n \iff \gcd(k, n) = 1$

Find all generators of $\mathbb{Z}_{36}, \mathbb{Z}_{48}$ } additively.
List all subgroups of $\mathbb{Z}_{36}, \mathbb{Z}_{48}$.

Q (13): Give two examples each of

1) finite cyclic groups, mention generators

2) infinite cyclic groups; mention generators.

Q (14): find all lattice of subgroups of $D_8, \mathbb{Z}_{24}, \mathbb{Z}_{36}$.

Q (15): ~~test~~ prove center of a group is a subgroup of G .

Q (16): let $f: \mathbb{Z}_8 \rightarrow \mathbb{Z}_4$

$f: \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{16}$

$f: (\mathbb{R}, +) \mapsto (\mathbb{R}^+, \cdot)$ by $f(x) = e^x \forall x \in \mathbb{R}$

$f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ by $f(x, y) = x + y$

be homomorphisms. Find kernels of each.

What are injective?

Q (17): Let $G = \mathbb{R} \times \mathbb{R}$. Describe G/H if

1) $H = \{(x, y) \mid y = -x\}$

2) $H = \{(x, y) \mid y = 2x\}$.

Q (18): Let $\alpha \neq 0$ be an element of a ring. prove that α can not be both a unit and a zero divisor.
Find all zero divisors and units in

1) $(\mathbb{Z}_{24}, +, \cdot)$

2) $(\mathbb{Z}_{36}, +, \cdot)$

3) $(S_3, \cdot)$, symmetric group of on 3 symbols.

3) 4) $(\mathbb{R} \times \mathbb{R}, +, \cdot)$, Cartesian plane

4) $(\mathbb{Z}_{17}, +, \cdot)$

5) $(\mathbb{Z}_p, +, \cdot)$ where p is prime.

Q (19): Let $\phi: \mathbb{R}[x] \rightarrow \mathbb{R}$ defined by $\phi(f(x)) = f(2)$
 $\forall f(x) \in \mathbb{R}[x]$. Find $\ker \phi$.
where $\mathbb{R}[x]$: ring of polynomials over $\mathbb{R}$

Q (20): Construct Quotient ring $\mathbb{Z}/6\mathbb{Z}$.
prove that multiplication is welldefined in $\mathbb{Z}/6\mathbb{Z}$.

Q (21): prove a field is an integral domain. Is converse true? Justify! when $\mathbb{Q}[\sqrt{2}]$ is a field? explain

Q (22): Explain in detail $\mathbb{Z}_3[x]/\langle x^2 \rangle$ is not a field.

Q (23): Reduce the cost

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1) $\alpha(x) + \langle x^2 \rangle$ in $\mathbb{Z}_3[x] / \langle x^2 \rangle$, where

$$\alpha(x) = x^{13} + 2x^{10} + 2x^5 + x + 2$$

~~2) $\alpha(x) + \langle x^2 \rangle$~~

Q (24): Let R be an integral domain; $f(x), g(x) \in R[x]$ with $f(x) \neq 0 \neq g(x)$. Show prove that $\deg(f(x) \cdot g(x)) = \deg f(x) + \deg g(x)$. Give a non-example for this!

Q (25): Consider the equation: $x^2 - 7x + 12 = 0$.
Find all solutions in $(\mathbb{Z}_{13}, +, \cdot)$ and $(\mathbb{Z}_{12}, +, \cdot)$.

Q (26): Determine if each polynomial is factorable or unfactorable.

1) $x^2 + 1$ in $\mathbb{Z}_7[x]$

2) $x^{1071} + 10x^{282} + 4x^{123} + 2$ in $\mathbb{Z}_{17}[x]$

3) $x^3 + x + 1$ in $\mathbb{Z}_5[x]$.

4) $x^{273} + 3x^{152} + 5x^{17} + 10$ in $\mathbb{Z}_{19}[x]$.

Q (27): Let F be a field, let $f(x) \in F[x]$ with $\deg f(x) \geq 2$.
If $f(x)$ has a root α , then $f(x)$ is factorable in $F[x]$.

Q (28): Ring R has cancellation property $\Leftrightarrow$ it has no zero divisors of zero.

Q (29): Define an extension field and algebraic element (or number), give example.

Q(30): (Group Theory!)

G is a group action $X \iff$ There exists a
homo. from $G \rightarrow G/G_x$ for each $x \in X$.

~~Def~~ Derive class equation.